

Yash Thube

Pune, India | thubeyash09@gmail.com | github.io

Experience

- **ML Research Engineer**, Future Sportler – Remote 01/2026 – Present
Working with multi-view geometry, biomechanics, kinematics, 3d reconstruction (human mesh recovery, gs, nerfs), temporal segmentation (handling occlusion in dynamic scenes), mot, pose estimation, sft/instruction tuning vlms, vision & diffusion models across different sports, camera angles, motion & physics.
- **Machine Learning Engineer**, Hudl India – Pune, MH 03/2024 – 12/2024
Worked on a multi camera yolo based pipeline for real-time player detection, integrating temporal tracking to handle occlusions across varied broadcast angles, finetuned vlms & video transformers on custom sports data for semantic action recognition, scaled deepsort mot to deliver 92% precision in crowded scenes, while mapping 2d coordinates to 3d spaces for precise kinematic localization.
- **Deep Learning Collaborator**, Omdena – remote, 10/2023 – 02/2024
Architected bert/roberta multimodal classifiers, applied rlhf via ppo for clinical safety alignment, and trained resnet backed cnn pipelines using mel-spectrograms for real-time vocal stress detection.
- **Computer Vision Collaborator**, AI Accelerator Institute – remote, 02/2023 – 06/2023
Worked on segmentation and self supervised learning for vision.

Research & applied interests

Broadly - 3D Vision & Embodied AI (Scene Understanding, Depth/Pose Estimation, 3D/4D Reconstruction, Multi-View Geometry), Representation Learning (SSL, 3DGS, Point Clouds, NeRFs), Model-Based RL (Predictive/Generative World Modeling, Spatial & Causal Reasoning)

Stack Used - PyTorch, OpenCV, Hugging Face (Transformers, Diffusers, PEFT, TRL), OpenMMLab (MMPose, MMDetection), PyTorch3D, Open3D, gsplat, Trimesh, SMPL-X, GVHMR, SAMURAI, Blender (BPY), MuJoCo, MuZero, EfficientZero, ByteTrack, BoT-SORT, OC-SORT, TrackNetV3, TransformerLens.

Projects & Research

- **MATS (arXiv preprint)** – A behavioral audit toolkit to detect pathological truth bias in VLMs, experiments include activation patching to causally localize failures in cross-attention layers and pooled representations across LLaVA, CLIP, and Qwen-VL architectures.
- **Paper Implementations** – PyTorch concept implementations of research papers, aimed at deepening my understanding of the underlying concepts.
- **Multimodal/VLMs Research Hub** – A technical resource for researchers exploring Vision-Language Models (VLMs) and multimodal learning, featuring seminal papers/models, datasets, benchmarks, ethical challenges, and research directions.

Github

Education

Savitribai Phule Pune University (SPPU)
B.E. Computer Engineering - 2026